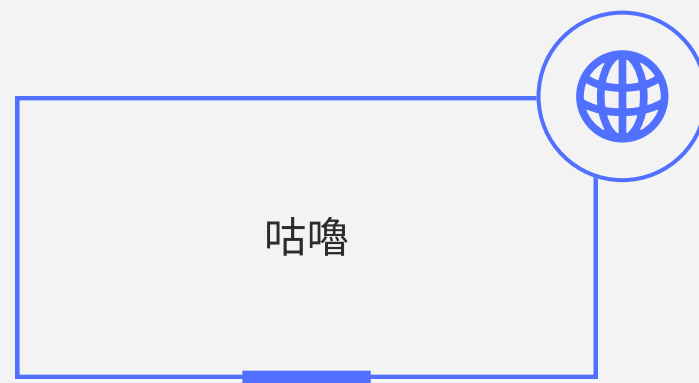
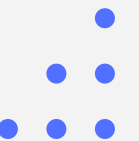
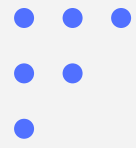


115-1 python 與 微控制器

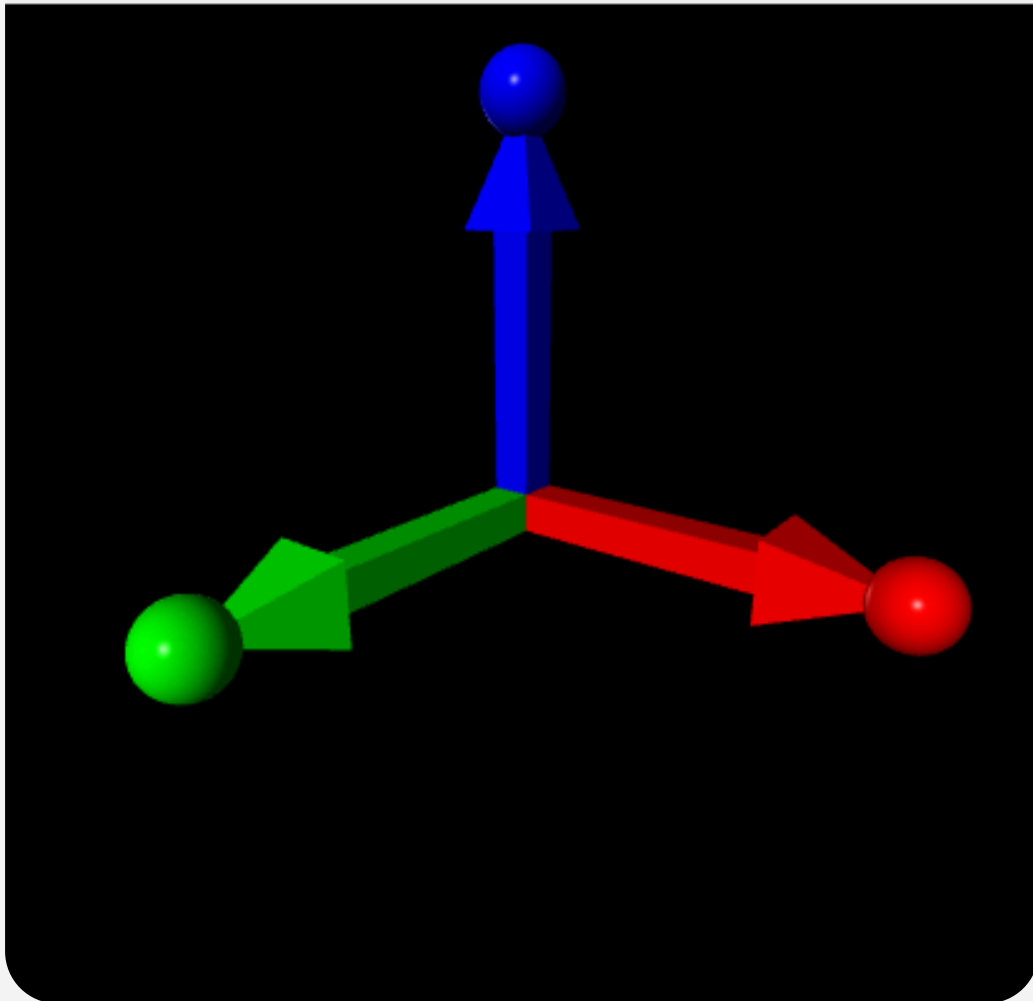


Presented by
高一丙18陶以恩





2026/09/11 09:10:29
ram Screenshot



```
1 Web VPython 3.2
2 |
3 a1 = arrow(pos=vec(0, 0, 0), axis=vec(1, 0, 0), color=color.red)
4 a2 = arrow(pos=vec(0, 0, 0), axis=vec(0, 1, 0), color=color.blue)
5 a3 = arrow(pos=vec(0, 0, 0), axis=vec(0, 0, 1), color=color.green)
6 b1 = sphere(pos=vec(1, 0, 0), radius=0.1, color=color.red)
7 b2 = sphere(pos=vec(0, 1, 0), radius=0.1, color=color.blue)
8 b3 = sphere(pos=vec(0, 0, 1), radius=0.1, color=color.green)
```

20260911-在
vpython 建立附件
箭头+球物件

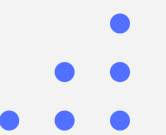


python

main.py

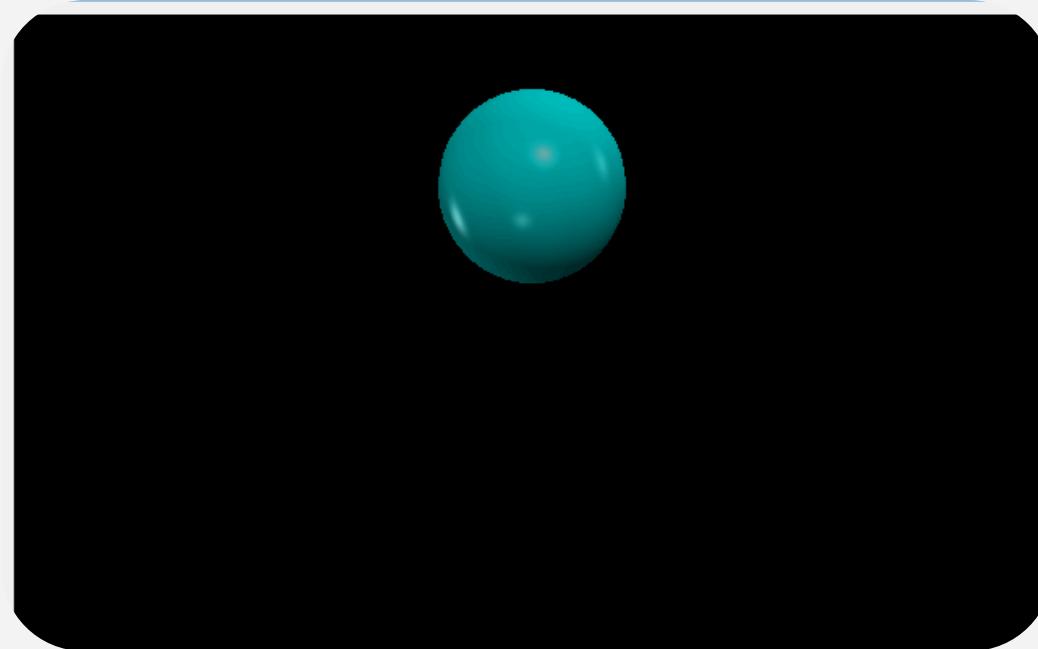
```
1 num = 5
2
3 while num >= 1:
4     print(f"Count: {num}")
5     num -= 1
6
7 print("發射！")
```

建立一個變數 `num = 5`，使用 `while` 迴圈讓程式從 5 倒數到 1（印出 `Count: 5`、`Count: 4` ...），迴圈結束後在最外層印出 "發射！"。





```
1 Web VPython 3.2
2
3 S1 = sphere(pos=vec(0, 0, 0), radius=2, color=color.cyan,opacity=
4
5 count = 0
6 while (count < 5):
7
8     count = count + 1
9
10
11
12 while (True):
13     rate(60)
14     S1.pos.y =S1.pos.y + 0.1
15     S1.radius =S1.radius + 0.01
16     S1.opacity =S1.opacity - 0.01
17
```



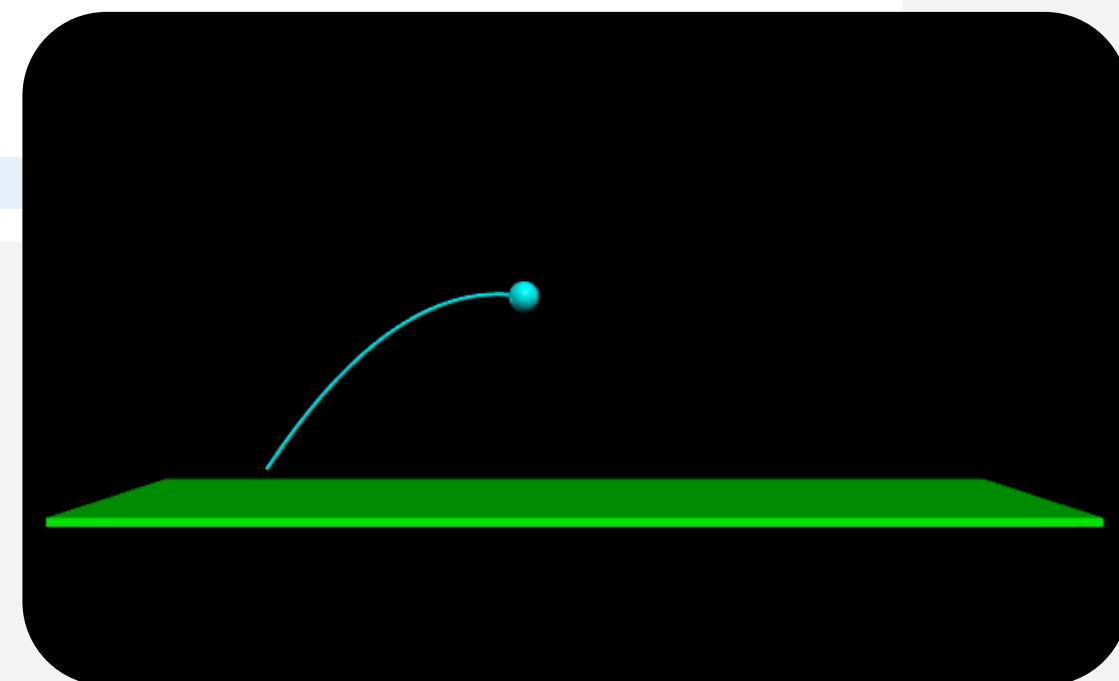
— 20260918- 讓一個
球球 向右動 然後消
失 --> 向 "右上" 動



python

```
1 Web VPython 3.2
2
3 S1 = sphere(pos=vec(-10, 10, 0), radius=2, color=color.cyan, opacity=1)
4
5 dt = 0.01
6 g = -10
7
8 vx = 5
9 vy = 0
10
11
12 while S1.pos.y > -10:
13     rate(100)
14
15     S1.pos.x = S1.pos.x + vx * dt
16     S1.pos.y = S1.pos.y + vy * dt
17
18     vy = vy + g * dt
```

20260918-鉛直上拋 改平拋





```
1 Web VPython 3.2
2
3 ground = box(pos=vec(0, -10, 0), size=vec(60, 0.5, 10), color=color.green)
4 S1 = sphere(pos=vec(-20, -8, 0), radius=1, color=color.cyan, make_trail=True, trail_type="curve")
5
6 dt = 0.01
7 g = -10
8
9 vx = 10
10 vy = 15
11
12 while S1.pos.y >= (ground.pos.y + S1.radius):
13     rate(100)
14
15     S1.pos.x = S1.pos.x + vx * dt
16     S1.pos.y = S1.pos.y + vy * dt
17
18     vy = vy + g * dt
```

— 20260918- 平抛
改斜抛



python

20261002- 箱内 弾性碰撞

```
1 Web VPython 3.2
2
3 scene.width = 1080
4 scene.height = 1080
5 scene.background = color.black
6
7 L = 5
8 t = 0.3
9 r = 0.3
10
11 wall_R = box(pos=vec( L, 0, 0), size=vec(t, 2*L, 2*L), color=color.red)
12 wall_L = box(pos=vec(-L, 0, 0), size=vec(t, 2*L, 2*L), color=color.red)
13 wall_T = box(pos=vec(0, L, 0), size=vec(2*L, t, 2*L), color=color.blue)
14 wall_B = box(pos=vec(0, -L, 0), size=vec(2*L, t, 2*L), color=color.blue)
15 wall_K = box(pos=vec(0, 0, -L), size=vec(2*L, 2*L, t), color=color.white)
16 wall_F = box(pos=vec(0, 0, L), size=vec(2*L, 2*L, t), color=color.white)
17
18 walls = [wall_R, wall_L, wall_T, wall_B, wall_K, wall_F]
19 normals = [vec(1,0,0), vec(-1,0,0), vec(0,1,0), vec(0,-1,0), vec(0,0,-1), vec(0,0,1)]
20
21 ball = sphere(pos=vec(0, 0, 0), radius=r, color=color.green,
22               make_trail=True, trail_color=color.green,
23               trail_radius=0.08, retain=200)
24 ball.v = vec(2.5, 3.2, -1.8)*2
25
26 running = True
27 def pause(b):
28     global running
29     running = not running
30
31 dt = 0.01
32 limit = L - t/2 - r
33
34 while True:
35     rate(100)
36
37     cam = scene.camera.pos
38     for i in range(6):
39         if dot(cam, normals[i]) > L:
40             walls[i].opacity = 0
41         else:
42             walls[i].opacity = 1
43
44     if running:
45         ball.pos = ball.pos + ball.v * dt
46
47         if abs(ball.pos.x) > limit:
48             ball.v.x = -ball.v.x
49         if abs(ball.pos.y) > limit:
50             ball.v.y = -ball.v.y
51         if abs(ball.pos.z) > limit:
52             ball.v.z = -ball.v.z
```

